

VERSION 1

The world is facing the climate change by carbon emission, so vehicles with renewable energy is invented, which is called EV. Everyone has seen EVs on the road, and UCSD have a lot of charging stations on campus. Every time you pay for the charging, do you ever think about why the rate is different among the stations and brands, and who you are exactly paying to?

We need to first know how electricity comes to your EV. After generation, electricity is first combined into California operator, then transmitted to local utility companies. In San Diego that is SDG&E, who plays as a retailer and then sell the electricity to commercial and industrial customers. UCSD buys energy as an individual from SDG&E for multiple purposes, like the building lighting, the HVAC system, and EV charging etc. To be noticed that although we drivers pay for energy charge with the unit as dollar per kWh, there's another charging cost: demand charge in the unit of dollar per kW. The purpose of demand charge is to limit the load demand, or otherwise the grid would be unstable and under the risk of overload.

So as UCSD aggregator, our target is quite simple: we wanna pay less to SDG&E and meet all the EV charging requirements. Let's take a look at the problem formulation. The math looks confusing, but the idea is quite easy to understand: minimizing the J subject to some EV physical constraints and SDG&E constraints. Equation 7 shows the total cost is composed of demand charge, energy charge and other charges. Among them demand charge is the dominance, which means if the total cost is 100 dollars, demand charge can be 90 dollars. Demand charge is actually applied to the highest load either during all daytime or during the peak hours which is 4pm to 9pm. So the most economical load should look like a plain, and it also prevents the grid from fluctuating, so minimizing the demand charge is a win-win situation for both customer and the grid. For every day every 15 minutes, we keep solving the minimization problem, and we will get the optimal profile of EV charging load to our best knowledge. This ideology is called model predictive control, namely update your decision after each observation.

However before we solve the problem, we need another important information: the forecast of EV arrival time, departure time, and their energy demand. Up to now, the charging stations do not necessarily require customers to provide their estimated AT and ED before charging. EV drivers will know that there's a brand called ChargePoint on campus, and they apply no control method, simply when an EV plugs in, the charger runs at the max power until it's unplugged or energy demand fully met. This method is called VOG in blue. We can do the forecast of EVs, predicting when they will come and what's their ED based on the historical sessions. Here I use some machine learning techniques like kNN and regression to drive the forecast model.

Now we can run the daily algorithm, and from the simulation graph it is clear that the methods with good forecast after optimization result in flat load profiles. VOG is the

worst with the huge spikes at the top, so the charging companies need to think twice about how they work on the energy schedules. To be noted that the orange one is the perfect forecast as the ideal case that can never be reached, and if a load is very close to it, the method is good. From the histograms we can also see the obvious cost reduction after applying the algorithms with the purple and brown.

In summary, with the combination of statistical forecast and a smart control method, UCSD can at the same reduce the payment to SDG&E for energy purchase, and make a contribution to stabilizing the grid. There is still a long way to go in saving the cost and improving the charging efficiency. Thank you.

VERSION 2

The world is facing a major challenge—climate change, driven by carbon emissions. To combat this, electric vehicles (EVs) powered by renewable energy have been invented. We see EVs everywhere now, and if you've been around UCSD, you've probably noticed plenty of charging stations. But have you ever wondered why charging rates differ between stations? More importantly, **who are you actually paying when you plug in your EV?**

Before your EV gets charged, electricity follows a long journey. It's first generated and sent to the **California power grid**, which is managed by the operator. From there, it flows to local utilities—in **San Diego, that's SDG&E**—which acts as a retailer, selling electricity to homes, and institutions like UCSD.

UCSD buys electricity from SDG&E for various purposes: **lighting buildings, running HVAC systems, and, of course, charging EVs.** But when it comes to paying for electricity, there's more to the bill than you may think.

Most EV drivers only think about the **energy charge**—the cost per kilowatt-hour. But there's another major factor: **the demand charge.** This charge is based on the highest peak power. Imagine if too many EVs come at once, the grid could become overloaded. **So the purpose of demand charge is to prevent sudden fluctuation in grid load.**

As an aggregator at UCSD, **our mission is simple: pay less to SDG&E while ensuring all EVs get charged.**

Let's take a quick look at the **optimization problem.** The math may look complicated, but the concept is straightforward: **we wanna minimize the total cost while following EV physical constraints and SDG&E's rules.**

- **Equation 7** breaks down the total cost into:
 - **Energy charge**
 - **Demand charge**
 - **Other minor charges**

Here's something shocking—if your total cost is **\$100, the demand charge alone could be \$90!** Therefore to reduce costs, we aim to **keep the charging load as flat as possible**—this not only saves money but also stabilizes the grid. **It's a win-win situation!**

But before we solve the problem, **we need a good forecast**—we need to predict:

1. **When EVs will arrive**
2. **How much energy they'll need**
3. **When they'll leave**

Right now, **UCSD's ChargePoint stations don't require drivers to input their estimated time.** Instead, as soon as an EV plugs in, it charges at max power until it's full or unplugged. **This simple charging method is called VOG.**

To improve this, **we use machine learning techniques like kNN and regression** to predict EV charging patterns based on historical data.

Now we can run our optimization daily, and the results are clear on the graph:

- **Uncontrolled VOG in blue causes large load spikes.**
- **The “perfect forecast” in orange is our ideal target—while we can't reach it, the closer we get, the better a method is!**
- **Optimized loads in purple and brown create flatter profiles.**

From the cost histograms, it's obvious—**optimal scheduling leads to significant cost reductions.**

In summary, by combining **statistical forecasting with smart control methods**, UCSD can **reduce its electricity bills** while **maintaining grid stability.** But there's more work to be done, and this is just the beginning of a smarter, greener future for EV

charging. **Thank you!** 🚗⚡